

iQ3R Grade 3 Mathematics

*Lesson 3 Module 1 Topic A*¹

*Instructor Version*²

June 10, 2022

This document describes the specific details for the third lesson of Module 1 in our adapted curriculum for 1:1 synchronous instruction in Grade 3 Mathematics³. Topic A is intended to help students develop increasing intuition about the meaning of multiplication. This is the final lesson in topic A.

¹ Adapted for individualized instruction from the EngageNY curriculum for New York State public and private instructors.

² Instructor notes will appear along the side column.

³ This is only an approximate estimate. Students will vary in skill level and preparedness

Student View: Slide 1

Lesson Overview

OBJECTIVE: Interpret the meaning of factors — the size of the group or the number of groups.

AGENDA

- Sprint
- Add to Multiply
- Arrays and Multiplication
- Number Bonds and Factors
- Problem Set
- Debrief

Lesson Overview

Standards

ADDRESSED: 3.OA.1, 3.OA.3

FOCUS: 3.OA.1 Interpret products of whole numbers, e.g., interpret 5×7 as the total number of objects in 5 groups of 7 objects each. For example, describe a context in which a total number of objects can be expressed as 5×7 .

OBJECTIVES: Interpret the meaning of factors—the size of the group or the number of groups. Lesson 3 solidifies students' ability to differentiate the meaning of factors. Students model dividing a whole into equal groups as well as analyze equal groups in scattered configurations and arrays to determine whether factors represent the number of groups or the size of groups. They create pictures, number bonds, and multiplication equations to model their understanding.

SUGGESTED LESSON STRUCTURE

- Fluency Practice — 6 minutes
- Application Problem — 4 minutes
- Concept Development — 20 minutes
- Student Debrief — 10 minutes

Student View: Slide 2

Fluency Practice

Sprint 1

A

Add Equal Groups

1.	$2 + 2 =$	
2.	2 twos =	
3.	$5 + 5 =$	
4.	2 fives =	
5.	$2 + 2 + 2 =$	
6.	3 twos =	
7.	$2 + 2 + 2 + 2 =$	
8.	4 twos =	
9.	$5 + 5 + 5 =$	
10.	3 fives =	
11.	$5 + 5 + 5 + 5 =$	
12.	4 fives =	
13.	2 fours =	
14.	$4 + 4 =$	
15.	2 threes =	
16.	$3 + 3 =$	
17.	2 sixes =	
18.	$6 + 6 =$	
19.	5 twos =	
20.	$2 + 2 + 2 + 2 + 2 =$	

Fluency Practice

Sprint 1

These Sprints are intended to reinforce material learned in Lesson 1. Students will be given 30 seconds to complete as many as possible. Then record the number correct and go on to Sprint 2. Students should be instructed to work as quickly as possible without sacrificing accuracy. They should try to improve on their score in the next sprint.

Student View: Slide 3

Fluency Practice

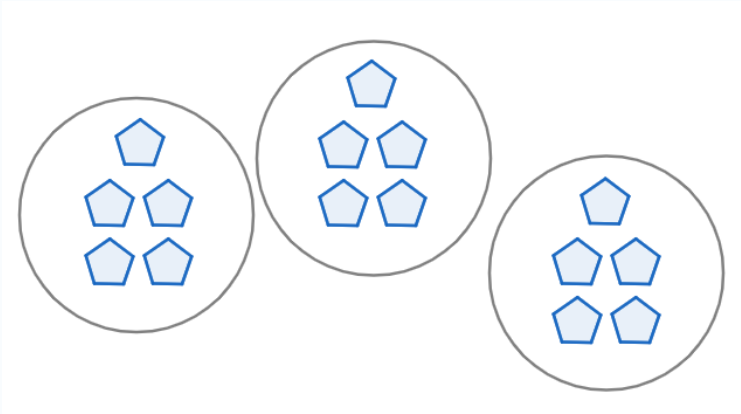
Sprint 2

B		
Add Equal Groups		
1.	5 + 5 =	
2.	2 fives =	
3.	2 + 2 =	
4.	2 twos =	
5.	5 + 5 + 5 =	
6.	3 fives =	
7.	5 + 5 + 5 + 5 =	
8.	4 fives =	
9.	2 + 2 + 2 =	
10.	3 twos =	
11.	2 + 2 + 2 + 2 =	
12.	4 twos =	
13.	2 threes =	
14.	3 + 3 =	
15.	2 sixes =	
16.	6 + 6 =	
17.	2 fours =	
18.	4 + 4 =	
19.	5 fives =	
20.	5 + 5 + 5 + 5 + 5 =	

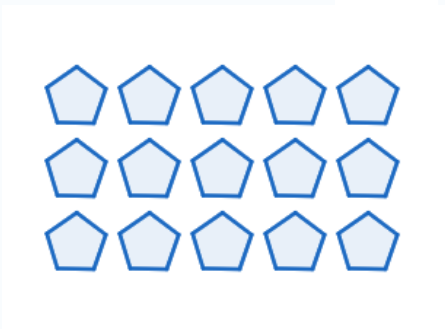
Fluency Practice

Sprint 2

Student View: Slide 4

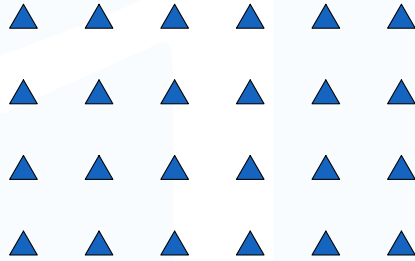
*Fluency Practice**Add to Multiply*

- How many groups are shown in the figure?
- How many items are in each group?
- Write an addition statement that describes the situation.
- Write a multiplication statement representing 3 fives equals 15.
- Can you arrange them in an array?

*Fluency Practice**Add to Multiply*

- This activity can be set up in Geogebra so that students can actually practice arrange them into the array shown below.
- Recall that in an array the “groups” for this picture are the rows.
- The number of things in each row is the number of things in each column. The array should be a 3 row 5 column display as shown in the figure.

Student View: Slide 5

*Concept Development**Arrays and Factors*

IN THE ARRAY above, the number of rows, and the number of triangles in each row are examples of *factors*. The total number of triangles is called the *product* of the factors.

- **Factor** — either the number of equal groups OR number of things in each group.
- **Product** — the total number of things all together.

PRACTICE: Use the array above to help you answer:

1. What factors are represented?
2. What product is represented?
3. What multiplication equation is represented?

*Concept Development**Arrays and Factors*

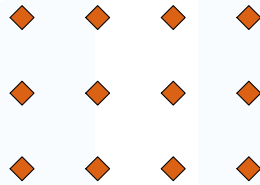
Using the array you will guide the student to the concepts of factors and products. It is very important that they have a concrete representation of the abstract concept of a factor. There are always at least two factors in a multiplication equation. The product is the result. Remember these are all just ways of counting.

- We have selected a fairly large array to illustrate the point that when collections get large, counting things one at a time is not going to work. Eventually we will learn multiplication facts. But these “facts” are simply rules for efficient counting.
- The factors are the number of rows and the number of columns. There are four rows and six columns so the factors are 4 and 6.
- The product is the total number of triangles. There are 24 triangles which can be discerned through a direct count.
- Therefore the array represents the multiplication equation $4 \times 6 = 24$.

Student View: Slide 6

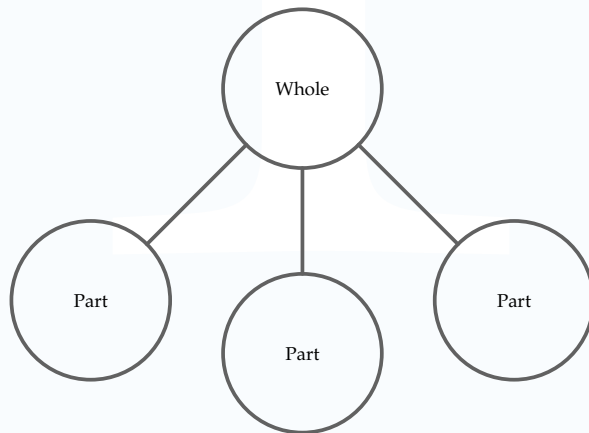
*Concept Development**Number Bonds*

ANOTHER REPRESENTATION you can use for arrays or multiplication problems is called a **number bond**. Consider the array



A number bond “bonds” the whole to its parts.

- whole = 12 red diamonds
- parts = 3 (rows)

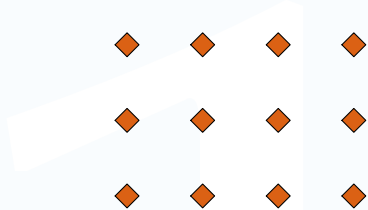
*Concept Development**Number Bonds*

- Number bonds relate a product to its factors. We represent the product as the whole and the number of things in each group as the part.
- The number of groups is represented by the number of circles in the lower part of the diagram. Emphasize the use of the terms “factor” and “product” as you move forward.
- Within each of these diagrams, stress that the numbers in the lower part of the diagram are used to “build” the number on top. In what cases will the number of circles (groups) be greater than the whole?
- In what cases will the numbers in the lower circles be equal to the whole?
- What numbers only allow a bond diagram with three circles?

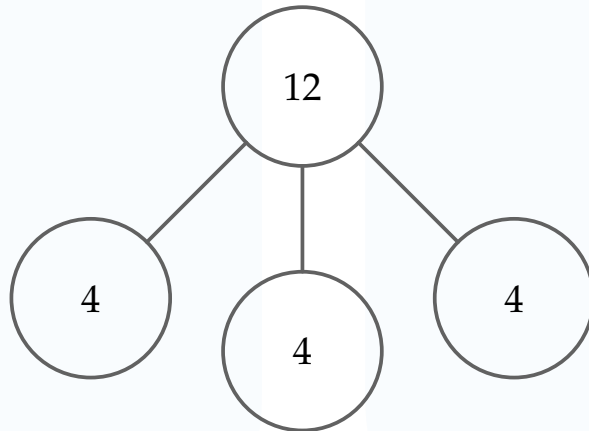
Student View: Slide 7

*Concept Development**Number Bonds*

This array



is represented by this number bond diagram.



- Three fours equal twelve.
- $4 + 4 + 4 = 12$
- $3 \times 4 = 12$
- 3 and 4 are factors of 12.

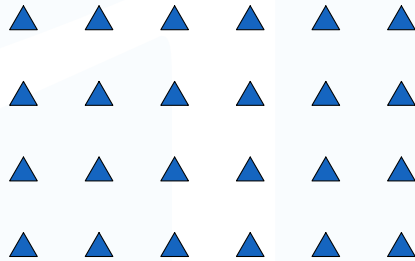
*Concept Development**Number Bonds*

Another good question to ask is how would the number bond diagram differ if we were talking about the multiplication $4 \times 3 = 12$? It would be a different picture, but it would represent the same number. It is important to emphasize this because it gets students ready to understand the commutative property of multiplication from several different representations.

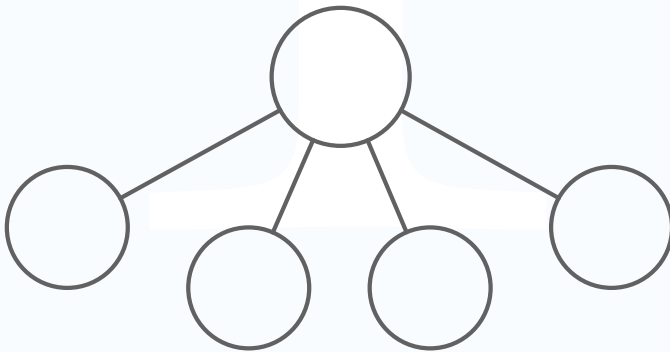
Student View: Slide 8

*Concept Development**Number Bonds*

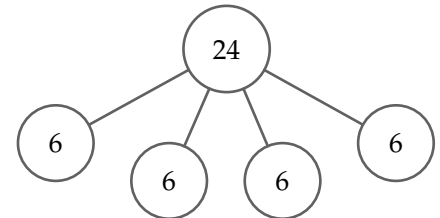
Fill in the number bond diagram that goes with this array.



- Four six's equal twenty four.
- $6 + 6 + 6 + 6 = 24$
- $4 \times 6 = 24$
- 4 and 6 are factors of 24.

*Concept Development**Number Bonds*

Solution:



Student View: Slide

Problem Set

Now you will work problems 1-6 of the lesson problem set.
(attached)

1

Problem Set

Name _____

Date _____

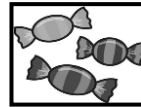
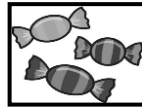
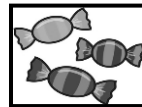
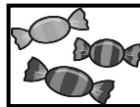
Solve Problems 1–4 using the pictures provided for each problem.

1. There are 5 flowers in each bunch. How many flowers are in 4 bunches?



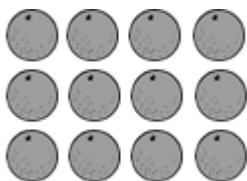
- a. Number of groups: _____ Size of each group: _____
- b. $4 \times 5 =$ _____
- c. There are _____ flowers altogether.

2. There are _____ candies in each box. How many candies are in 6 boxes?



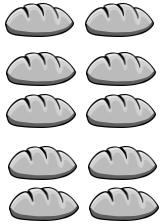
- a. Number of groups: _____ Size of each group: _____
- b. $6 \times$ _____ $=$ _____
- c. There are _____ candies altogether.

3. There are 4 oranges in each row. How many oranges are there in _____ rows?



- a. Number of rows: _____ Size of each row: _____
- b. _____ $\times 4 =$ _____
- c. There are _____ oranges altogether.

4. There are _____ loaves of bread in each row. How many loaves of bread are there in 5 rows?



a. Number of rows: _____ Size of each row: _____

b. _____ $\times$ _____ = _____

c. There are _____ loaves of bread altogether.

5. a. Write a multiplication equation for the array shown below.

X X X
X X X
X X X
X X X

b. Draw a number bond for the array where each part represents the amount in one row.

6. Draw an array using factors 2 and 3. Then, show a number bond where each part represents the amount in one row.

*iQ3R Mathematics — Geometry*¹

*Lesson 6 Module 1*²

June 11, 2022

This sixth lesson is a bit of a departure from the New York curriculum. Instead of starting into Topic B, we find it best to review the material in Topic A and to tie together some concepts we left unaddressed in Topic A. Students will still practice using the construction tools and techniques introduced in lessons 1 through 5, but here we will close the gaps in some of the reasoning that was used to assert some of the claims made in those lessons.

Student View: Slide 1

Overview

IN THIS LESSON you will bring together all the ideas you learned in lessons 1 through 5. This will complete this unit on constructions. Specifically the aim of this lesson is to

1. Review basic constructions: you will also learn to find perpendiculars through points on lines; parallels through points not on lines.
2. Discuss postulates vs. assumptions.
3. Introduce measurement concepts: how to measure distance between various geometric objects.
4. Introduce you to symmetry.
5. Help you see “why” the concurrences we saw in lessons 4 and 5 occur.

¹ Typically in the United States grade 9 students take either Algebra 1 or Geometry depending on the school and the performance of the student in 8th grade. Generally, Geometry is taken by students who are on a more accelerated track.

² Adapted for individualized instruction from the EngageNY curriculum for New York State public and private instructors.

Overview

Lesson 6 Supported Standards

Common Core Focus Standards

G-CO.A.1

Know precise definitions of angle, circle, perpendicular line, parallel line, and line segment, based on the undefined notions of point, line, distance along a line, and distance around a circular arc.

G-CO.D.12

Make formal geometric constructions with a variety of tools and methods (compass and straightedge, string, reflective devices, paper folding, dynamic geometric software, etc.). Copying a segment; copying an angle; bisecting a segment; bisecting an angle; constructing perpendicular lines, including the perpendicular bisector of a line segment; and constructing a line parallel to a given line through a point not on the line.

G-CO.D.13

Construct an equilateral triangle, a square, and a regular hexagon inscribed in a circle.

Student View: Slide 2

*Review with Applications**Basic Constructions*

You now know¹ three basic constructions using the drawing tools

- Given a line segment, find the perpendicular bisector to the line segment.
- Given an angle, find the angle bisector of that angle.
- Given an angle, copy the angle along another segment or ray.

Using these basic constructions, you now are able to

- Construct an equilateral triangle
- Construct a regular hexagon
- Find the perpendicular to a given line which contains a specified point not on the line.
- Construct the incenter and circumcenter of a triangle as well as the corresponding circles.

CHECK FOR UNDERSTANDING: Given a triangle $\triangle ABC$ draw

1. The circumcircle
2. The incircle

CHALLENGE: When will the circumcenter of a triangle be on the exterior of the triangle? When will it be on one of the sides of a triangle?

*Review with Applications**Basic Constructions*

1. Students have seen these procedures and have been using them over the last five lessons so the procedures will not be reproduced here. Be prepared to verbally review them if necessary.

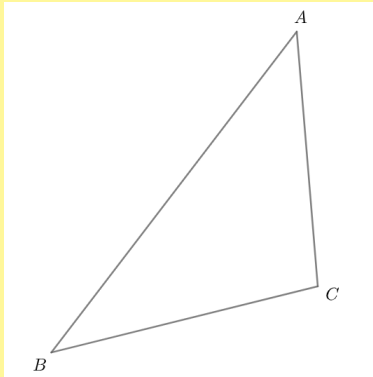
Do we want to have some form of digital handout they can reference?

Lesson Design Specifications (Please Read)
Guided Practice:

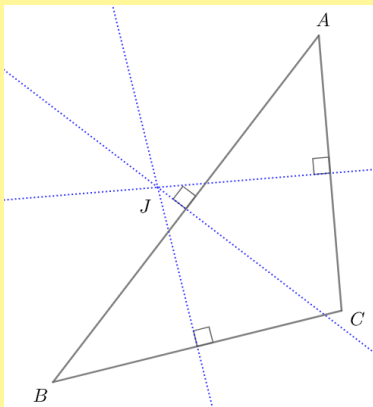
At this point student and teacher will do one of the following:

- Navigate to Geogebra on a shared screen **OR**
- Launch an app that opens up a shared window for interactive geometry **OR**
- Share a whiteboard with appropriate drawing tools.

Instructor will now use the following sequence on first a scalene then an obtuse triangle. We show only the case for the obtuse.

CONSTRUCTION SEQUENCE

STEP 1

Find the perpendicular bisectors of the segments of $\triangle ABC$. Label their point of concurrency¹ J .


STEP 2

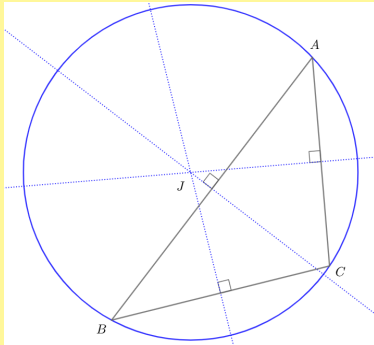
The result should look something like the figure shown². Ask, "What radius should you use for the circle centered at J ?" *Answer:* Any of the equal distances AJ , BJ or CJ will do.

Now draw circle J with the appropriate radius.

1. Ask, "Do we really need to find all three perpendicular bisectors to draw the circumcircle?"
2. Note that we have not included **arrowheads** in the picture. When the line extends to the border of the graphic, that also is intended to represent a line. When we want to emphasize that a portion of a line is being viewed, we will do that by keeping the entire structure within the viewing window and using the arrowheads where the segment terminates.

Lesson Design Specifications (Please Read)

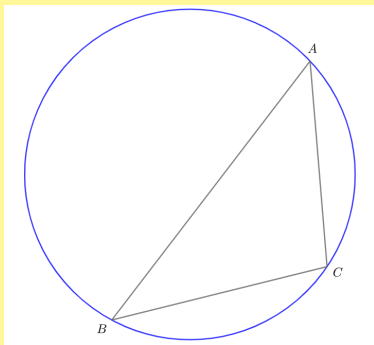
Guided Practice: Continued



STEP 3

The result should look something like the figure shown. Now clear away the auxillary lines and miscellaneous label J .

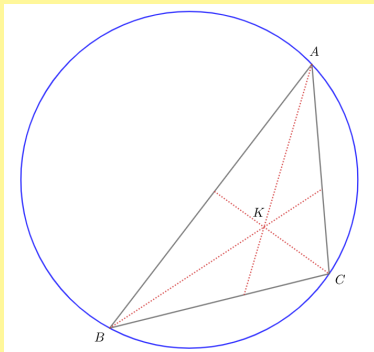
Leave the circum-circle in view.



STEP 4

Now draw the bisectors of each of the angles of triangle $\triangle ABC$.

Label the incenter as K .



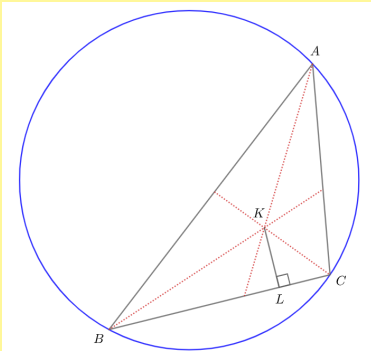
STEP 5

The result should appear somewhat like what you see here in this figure.

Next, draw a perpendicular segment from K to side BC and label its intersection with BC as L .

Lesson Design Specifications (Please Read)

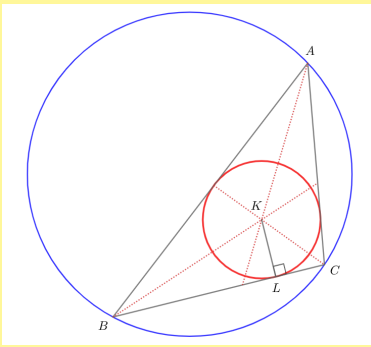
Guided Practice: Continued



STEP 6

The result should appear somewhat like what you see here in this figure.

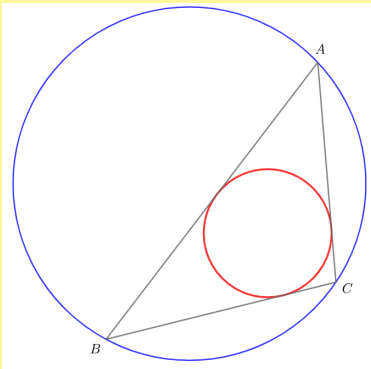
Finally, draw circle K centered at K with radius KL .



STEP 7

The result should appear somewhat like what you see here in this figure.

Now eliminate all the superfluous lines and labels.



The final figure with a triangle showing BOTH the incircle and the circumcircle.

Now we return to the main instructional sequence.

Student View: Slide 3

*Review with Applications**Extending the Results*

Previously: For given line k and point P not on k we saw how to use our basic constructions to find the line perpendicular to k through P . We now make a definition/notation which we will stay with for the course.

DEFINITION $k^\perp(P)$: Given line k and point P (either on or off of k), we represent *the unique* line which is perpendicular to k and contains P by the notation¹ $k^\perp(P)$.

DEFINITION $k^\parallel(P)$: Given a line k and a point P (again, either on or off of k), we represent the unique line which is parallel to k through P by the notation $k^\parallel(P)$. If a point P is on k we take $k^\parallel(P)$ to simply be k .

KNOWLEDGE POINT: Using the basic constructions you now can

1. Construct $k^\perp(P)$, given line k and point P in any case — whether P is on k or not.
2. Construct $k^\parallel(P)$ for any given line k and point P .

Let's work out the details of (1) first.

*Review with Applications**Extending the Results*

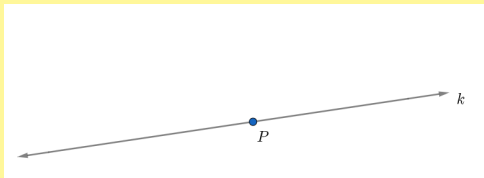
1. The function notation is intentional. For each line k in the plane $k^\perp$ defines a function from the set of points in the plane to the set of lines in the plane. This function has interesting properties on its own.

Lesson Design Specifications (Please Read)

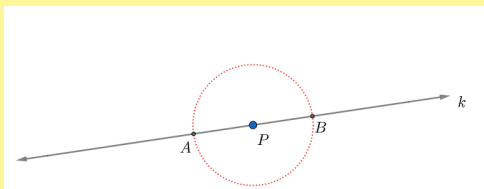
Guided Practice 2: Construction of $k^\perp(P)$; case P is on k .

At this point student and teacher will do one of the following:

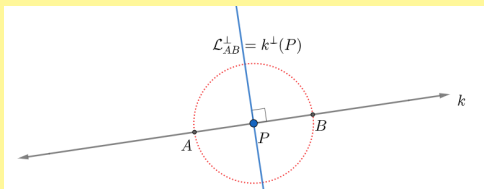
- Navigate to Geogebra on a shared screen **OR**
- Launch an app that opens up a shared window for interactive geometry **OR**
- Share a whiteboard with appropriate drawing tools.



STEP 1: Fix line k with selection of arbitrary point P on k . Draw a circle with center at P and label intersections as A and B .



STEP 2: Now find $\mathcal{L}_{AB}^\perp$ — the perpendicular bisector of segment AB .



STEP 3: Done. We have done what we set out to do. We have found $k^\perp(P)$ — the line perpendicular to k through the point P *located on k* .

We now return to the main sequence of instructional slides.

Student View: Slide 4

*Review with Applications**Construction of $k^\perp(P)$.*

NOTATION: If a line k is specified with selected point P on k , we will represent the line perpendicular to k containing the point P by the symbol $k^\perp(P)$ — pronounced “k perp of P”.

KNOWLEDGE POINT: Construction of $k^\perp(P)$ from line k and point P on line k .

- Step 1: Locate P on line k .
- Step 2: Draw circle P centered at P with any positive radius r . Since $r > 0$ circle P meets k in exactly two points which we label A and B .
- Step 3: Draw $\mathcal{L}_{AB}^\perp$ — the perpendicular bisector of segment AB .
- Observation: $\mathcal{L}_{AB}^\perp = k^\perp(P)$.

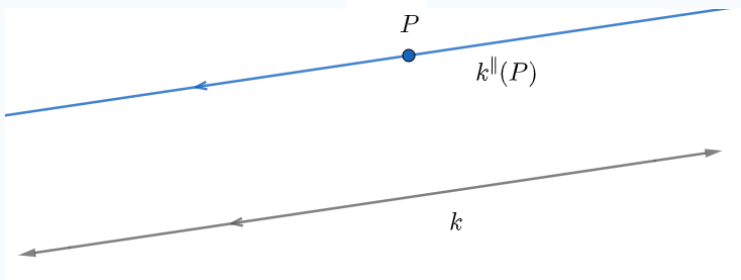
We now address the second problem we considered.

*Review with Applications**Construction of $k^\perp(P)$.*

Student View: Slide 5

*Review with Applications**Construction of $k^{\parallel}(P)$*

IMPORTANT PROBLEM: Given line k and point P , off k , find a line which is both parallel to k and contains P . Recall that we use the notation¹ $k^{\parallel}(P)$ to represent this line: **The line *parallel* to line k containing the point P which is off k .** This is illustrated in the figure below.



We use the arrows² along the lines for emphasis that these lines are parallel.

Let's see how this construction works 🧐

*Review with Applications**Construction of $k^{\parallel}_P$*

1. Be sure to emphasize that the words go with the symbol. When they see the symbol $k^{\parallel}(P)$ they should be thinking "the line parallel to line k through the point P ." We use the symbol so we don't have to keep writing the cumbersome phrase over and over again, but we **MUST** remember that the symbol is encoding the language of geometry.
2. We mean the arrows around the "center" of the figure. Recall our convention: lines are either represented as in the figure by k using arrows on the ends of the segment, or as in the figure by $k^{\parallel}(P)$ which simply runs from one side of the viewing window to the other.

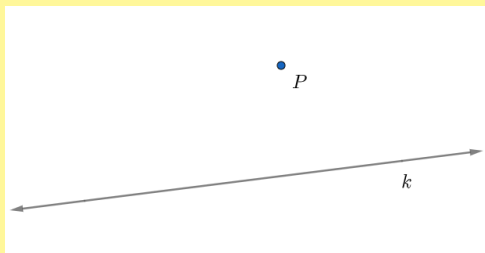
Lesson Design Specifications (Please Read)

Guided Practice:

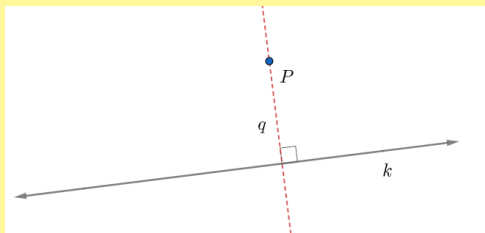
At this point student and teacher will do one of the following:

- Navigate to Geogebra on a shared screen **OR**
- Launch an app that opens up a shared window for interactive geometry **OR**
- Share a whiteboard with appropriate drawing tools.

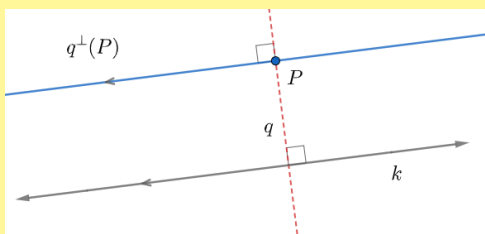
Instructor will now guide the student through the steps required to construct $k^{\parallel}(P)$.



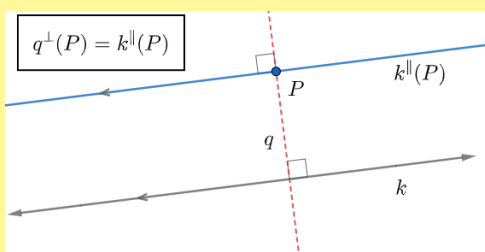
STEP 1: Start with a given line k and point not¹ on k which we have labeled P .² Using the ideas from lesson 4, construct the perpendicular to k containing point P . Call this q . (Observe that $q = k^{\perp}(P)$.)



STEP 2: You should get a figure similar to the one shown here at the left. Now have the student construct $q^{\perp}(P)$ — the line perpendicular to q through P .



STEP 3: That is it! Amazing! Actually $q^{\perp}(P)$ is what you are looking for. We emphasize this in the next figure.



ALL DONE! QED³ 🧐

1. What would happen if we tried to pick point P on line k ?
2. Again, stress that what you label the lines and points does not matter. Notice that when we label lines other than by using notation like $\overleftrightarrow{AB}$, we use either lower case letters or the more elaborate $\mathcal{L}$.
3. QED — *quod erat demonstrandum* — Latin for “what was to be demonstrated”

Student View: Slide 6

*Review with Applications**Construction of $k^{\parallel}(P)$*

KNOWLEDGE POINT: If a line k and a point P not on¹ k are specified, then we use the following procedure to find $k^{\parallel}(P)$ — the line parallel to k through the point P .

- Step 1: Draw line $k^{\perp}(P)$ containing point P .
- Step 2: Draw the perpendicular through P to line $k^{\perp}(P)$.
- Step 3: Verify that in fact the following beautiful equation² holds!

$$k^{\parallel}(P) = (k^{\perp}(P))^{\perp}(P)$$

OBSERVATION: This equation of line descriptors is an extremely concise way of saying that, "Two lines perpendicular to the same line are parallel to each other."

CHECK FOR UNDERSTANDING: Using our notation³, what should be the result of finding $(k^{\parallel}(P))^{\perp}(P)$?

*Review with Applications**Construction of $k^{\parallel}(P)$*

1. Recall that if P is on K then $k^{\parallel}(P)$ is just k .
2. This is truly an instance of where the choice of effective notation really does convey more information than expected. The language here that goes with this little equation is, "The perp *through* P on the perp *from* P to k is the *parallel* to k through P ."
3. Focus on the language. The expression says "find the perpendicular through P , to the line *parallel to* k which contains P . This must be the perpendicular through P to k . So the answer is simply $k^{\perp}(P)$."

Student View: Slide 7

*Concept Development**Postulates****Parallel Postulate:***

Given a point P and line k not containing P there exists *exactly one* line containing P which is parallel to k .

Assumptions in Geometry are actually called “postulates”.

- An assumption can be easily changed.
- A postulate¹ is something that is taken to be true and is not easily changed.
- This “parallel postulate” changed the world of geometry!
- Since it is an assumption and not a “fact”, people wondered if it was a necessary condition² of the assumed “Euclidean Assumptions.”
- This postulate actually characterizes what we call “flat space” and it is not necessarily true for all “spaces.”³

*Concept Development**Postulates*

1. Postulates are the foundation of the theory of geometry. We can change the parallel postulate to read that there are infinitely many lines parallel to k . When we do that we change from “flat” geometry to non-Euclidean geometry.
2. It is not a necessary condition of the assumptions of Euclidean Geometry. This is in fact the starting point for non-Euclidean geometries. For centuries, mathematicians tried to “prove” the parallel postulate as a consequence of other assumed “facts” and were miserably thwarted. Now we know that this postulate is actually what characterizes the Euclidean “flat” space.
3. The physical universe is generally considered to be a 4-dimensional space-time manifold that is “curved” due to the presence of matter which is equivalent to gravity according to Einstein.

Student View: Slide 8

Concept Development

Postulates

The Ruler Postulate:

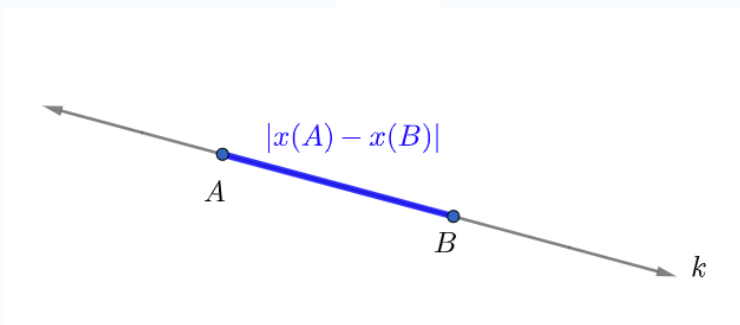
Let point O and P be chosen on line k . Let the real number system, $\mathbb{R}$, represent measurements in a fixed system of units.¹ Then there exists a one to one² correspondence $x : k \rightarrow \mathbb{R}$, such that $x(O) = 0$, $x(P) = 1$, and for all points A and B on k ,

$$|x(A) - x(B)| = AB.$$

Moreover, given any real number ζ ,³ there exists exactly one point Z on k , such that $x(Z) = \zeta$.

The mapping x is called a *coordinate system* on k . It is also referred to as a *coordinationization* of k .

So the content of the ruler postulate is that any line can be coordinatized. Coordinationization of a line is far from unique. Indeed, if x and y are both (linear)⁴ coordinate systems on k , a previous challenge problem showed that $y = mx + b$ for some constants m , and b .



Concept Development

Postulates

1. For example, it could be units of time, money, length and so on. The idea is that scalars are not just numbers when it comes to measurement in geometry. We measure lengths, or intervals, in units in some fixed system. Be sure to emphasize this. In geometry when we say, for example $AB = 2$, we most always mean 2 of "something" — unless you are working in a highly specialized dimensionless framework which is way beyond the scope of this course.
2. One-to-one means that if $x(A) = x(B)$, then $A = B$. The mapping is also surjective (onto). This is the content of the second part of the postulate.
3. This is the Greek letter "zeta".
4. Curvilinear coordinate systems are also possible: a standard coordinate system on the "postive" axis is given by $x = \log t$. This is the system in which earthquake magnitudes are measured.

Student View: Slide 9

Concept Development

Point Plotting Theorem

From the Ruler Postulate we immediately get the important

Point Plotting Theorem: Let $\overrightarrow{OQ}$ be any given ray. Let $r > 0$ be given in a specified unit system for measurements on $\overrightarrow{OQ}$. Then there exists a unique point P on $\overrightarrow{OQ}$ such that $OP = r$.

We will be studying the idea of proof very soon. Here we will demonstrate a little bit about what is meant by “proof”. A proof in mathematics is simply a reasoned argument based on accepted assumptions.

What have we to assume here? **The Ruler Postulate:** From this we get a coordinate system “ x ” on line OQ with $x(O) = 0$ and $x(Q) = 1$. Hence $\overrightarrow{OQ}$ is the set of points with “positive” coordinates in the “ x ” system.¹

Take $\zeta = r$ in the second part of the Ruler Postulate. The postulate assures us the existence of a point $Z = P$ with $x(P) = r$. Since

$$OP = |x(P) - x(O)| = |x(P) - 0| = |x(P)| = x(P) = r,$$

we have demonstrated what we desired to show².

CHECK FOR UNDERSTANDING: Will x composed with $t \mapsto t^2$ still give us these results? ³

Concept Development

Point Plotting Theorem

1. A coordinatization is exactly that ... it is a system for assigning numbers to points and conversely. Emphasize that this is entirely up to the experimenter. But once a coordinatization has been selected, all measurements must be agreed on within that coordinate system. Be very careful if you have to “change coordinates”. Make sure to keep track of the bookkeeping if you do that. This is especially important in Physics.
2. Well, we have really only shown “existence” ... can you explain why uniqueness also follows? (Answer: If P' is any other point on ray OQ with $OP' = r$, then $x(P) = x(P') = r$, and by injectivity of x we have that $P = P'$.)
3. Yes! Since $t \mapsto t^2$ is one to one on the positive axis, take $\zeta = \sqrt{r}$ and apply the Ruler Postulate.

Student View: Slide 10

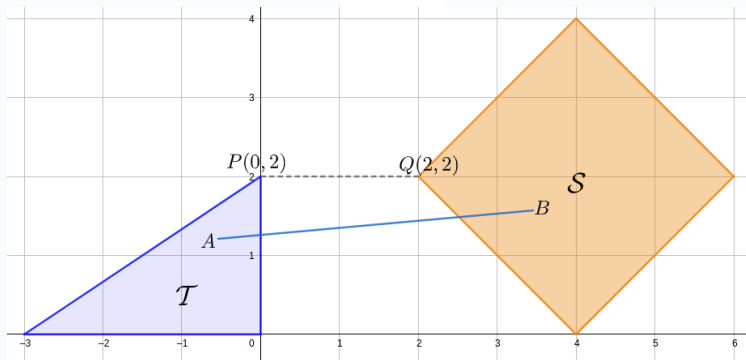
Concept Development: Measurement

Distance

KNOWLEDGE POINT: *Distance Between Points* Let a system of units be selected for measurements of distance between points on ANY line. If x is a coordinate system on line AB then the distance between A and B is $|x(A) - x(B)|$.

We define distance between “extended objects”¹ in terms of the distance between points.

EXAMPLE: What is the distance between the triangle and the square in the figure below?



Since $PQ = 2$ and every other segment $\overline{AB}$ with A in T and B in S satisfies $AB > PQ = 2$, we say that the distance between S and T is simply $PQ = 2$.

DEFINITION: *Distance Between Sets of Points*

Let T and S be sets of points in the plane (or space). We define the *distance between* T and S , denoted $\text{dist}(T, S)$ to be the *minimum*² of all lengths AB where A lies in T and B lies in S .

Concept Development: Measurement

Distance

1. Extended objects are objects which are composed of possibly more than one point. The idea here is to use our notion of distance between points to develop the concept of distance between objects.
2. In actuality, we should have

$$\text{dist}(T, S) = \inf\{AB \mid A \in T, B \in S\}$$

The definition given here works fine when the sets are compact. It is certainly true for simple plane figures like polygons. This is only a course in elementary geometry. So this definition will do well for us. Interested and thoughtful students will do well to consider the shortcomings of this definition. For example, the min does not exist between the point $(2, 4)$ and the *open* unit disc.

Student View: Slide 11

*Concept Development: Measurement**Distance*

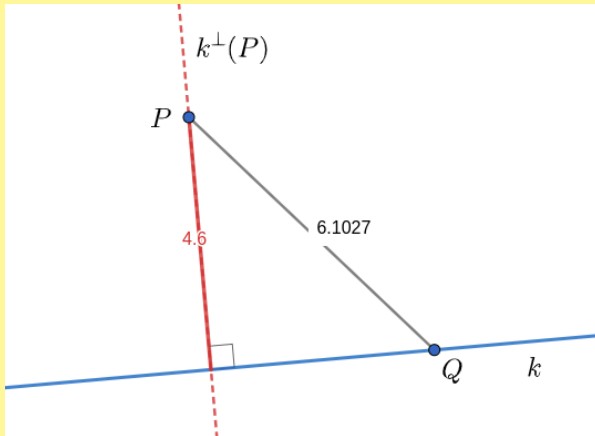
CHECK FOR UNDERSTANDING: Use our generalized definition of *Distance Between Sets of Points* to verify that the right definition of distance from a point to a line is measured along a perpendicular¹.

Lesson Design Specifications (Please Read)*Guided Practice:*

At this point student and teacher will do one of the following:

- Navigate to Geogebra on a shared screen **OR**
- Launch an app that opens up a shared window for interactive geometry **OR**
- Share a whiteboard with appropriate drawing tools.

Here the Instructor will guide the student to use the following app² which allows the student to move point Q along k and see the distance PQ continuously update. They will see that PQ takes its “min” exactly when Q takes on the location of $k^\perp(P) \cap k$.

*Concept Development: Measurement**Distance*

1. Let P be a point selected off of a given line k . You will want to help the student identify the two sets $T = \{P\}$ and $S = k$. The exercise consists in showing that given any arbitrary point Q on k the distance PQ is minimized when Q is on $k^\perp(P) \cap k$.
2. This app is very easy to use. Students will see at once that the only way to get the smallest value of PQ is to select that point Q both on k and on the perpendicular $k^\perp(P)$.

Student View: Slide 12

*Concept Development**Symmetry*

DEFINITION: **Reflection in a Line:** Let line k in the plane be given. A **reflection** in k is a function R_k from the plane to itself which has the following properties:

1. For all P not on k , $\overline{R_k(P)P} \perp k$.
2. For all P on k , $R_k(P) = P$.
3. k is the perpendicular bisector of $\overline{R_k(P)P}$ for all points P not on k .

*Concept Development**Symmetry*

1. This says that segments formed from images and their pre-images are always perpendicular to the line of reflection.
2. This says that the line of reflection is fixed under the transformation.
3. This says that k contains the mid-points of each of the segments $\overline{R_k(P)P}$.

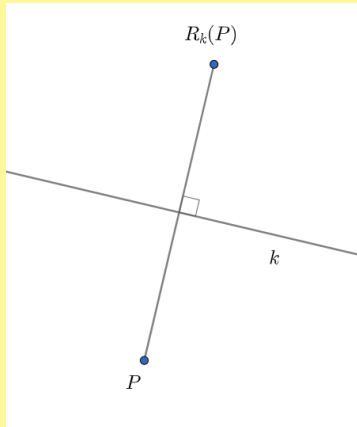
Lesson Design Specifications (Please Read)*Guided Practice:*

At this point student and teacher will do one of the following:

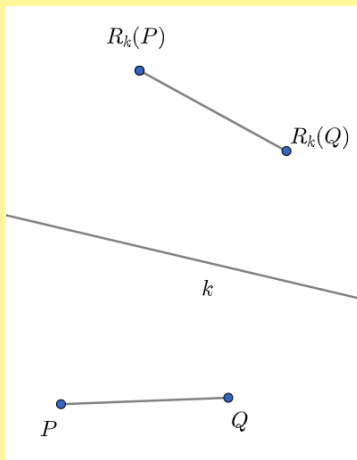
- Navigate to Geogebra on a shared screen **OR**
- Launch an app that opens up a shared window for interactive geometry **OR**
- Share a whiteboard with appropriate drawing tools.

This sequence of demonstrations will help our students understand the idea of “reflection in a line” as a rigid transformation of the plane. This is a huge concept coming up. We get a brief glimpse of it here.

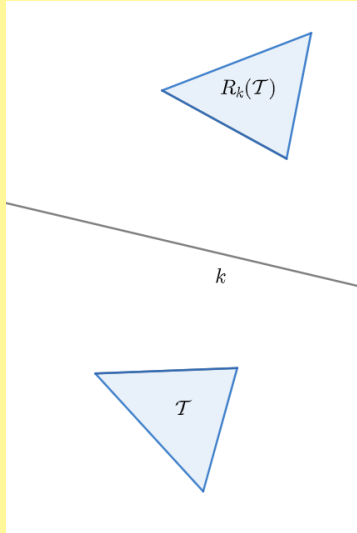
1. Reflection of point P in line k .



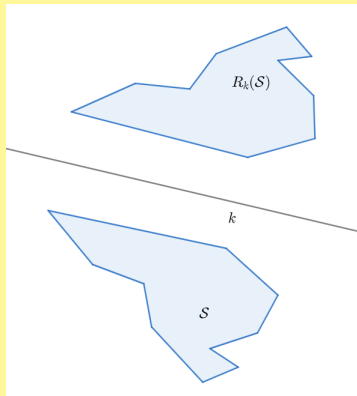
2. Reflection of a segment PQ in line k .



3. Reflection of a triangle in line k .



4. Reflection of a more complicated figure about line k .



We now return to the main instruction sequence.

Student View: Slide 13

(HUGE¹) KNOWLEDGE POINT: A figure $\mathcal{K}$ is *symmetric* with respect to a line k if $R_k(\mathcal{K}) = \mathcal{K}$.

SYMMETRIES OF IMPORTANT FIGURES:

1. Let Q be a square and let d be a given diagonal. Then Q is symmetric about d .
2. Let d be any diameter of any circle. The circle is symmetric about d .
3. Let $\overline{AB}$ be any given segment. Then $\overline{AB}$ is symmetric with respect to its perpendicular bisector $\mathcal{L}_{AB}^\perp$.²
4. Let angle $\angle ABC$ be given. Let $\overrightarrow{BK}$ be the angle bisector of $\angle ABC$. Then $\angle ABC$ is symmetric with respect to $\overrightarrow{BK}$.³

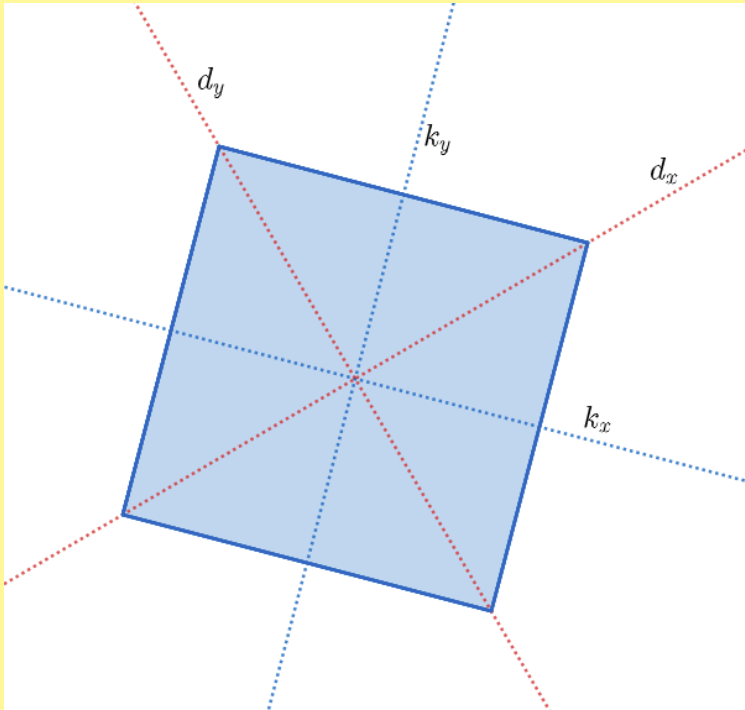
Lesson Design Specifications (Please Read)

Guided Practice: Demonstrations of Symmetries

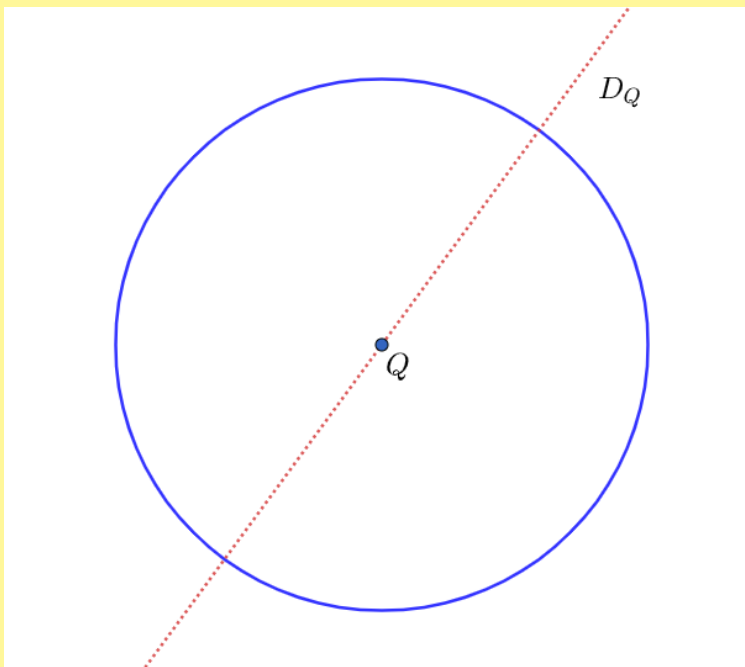
At this point student and teacher will do one of the following:

- Navigate to Geogebra on a shared screen **OR**
 - Launch an app that opens up a shared window for interactive geometry **OR**
 - Share a whiteboard with appropriate drawing tools.
1. Use the reflection tool to show that the square is invariant under reflection symmetries in each of the lines k_x , k_y , d_x , and d_y .

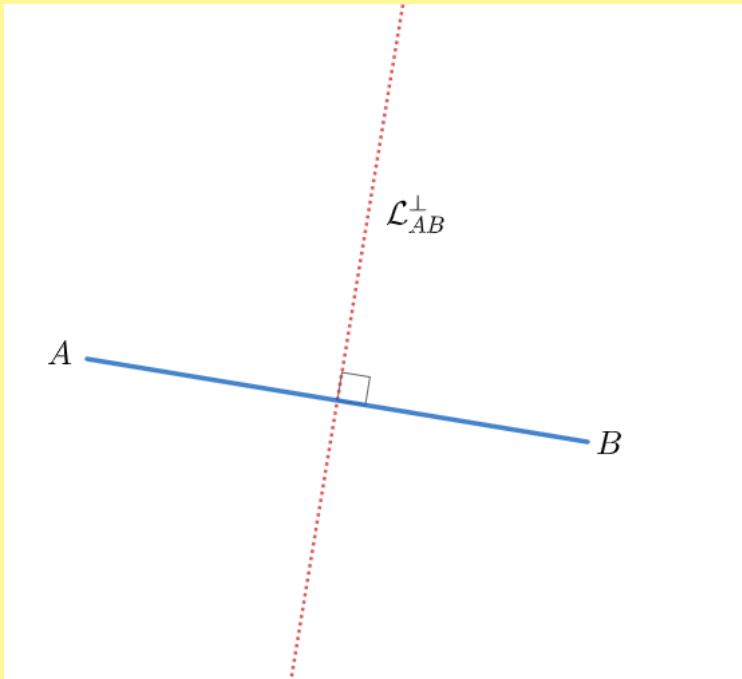
1. It cannot be overstated how important this concept is. The idea of symmetry in mathematics is ubiquitous. Later we will investigate rotational symmetry and symmetry via reflection in points. Figures in the plane can be described in large measure by their symmetry groups. For example, a circle is characterized by rotational invariance about its center through every angle, and reflection invariance about every line containing its center. It is also properties of symmetry that inform our ideas of probability; since a coin is “symmetric” about its axes of rotation, we should expect a 50:50 chance of getting heads or tails.
2. It is this fact that allows us to conclude WHY the perpendicular bisectors of the sides of a triangle are concurrent.
3. It is this fact that allows us to conclude WHY the angle bisectors of a triangle are concurrent.



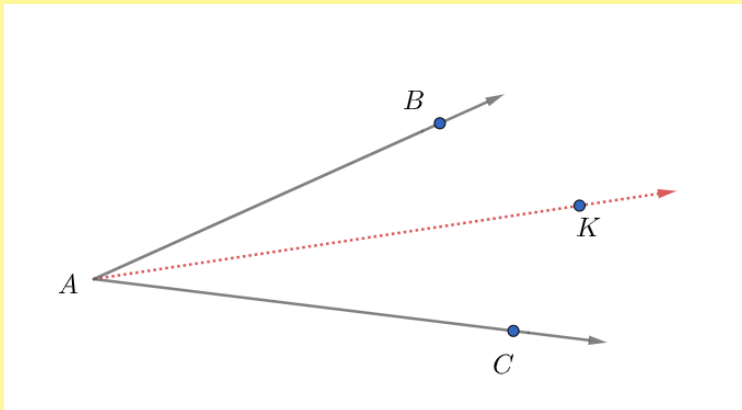
2. Use the reflection tool to show that a circle is reflection invariant through any line containing a diameter. Any line D_Q containing the center of a circle Q contains its diameter as shown.



3. Use the reflection tool to show that a segment $\overline{AB}$ is reflection invariant about its perpendicular bisector $\mathcal{L}_{AB}^{\perp}$.



4. Use the reflection tool to show that an angle $\angle BAC$ is reflection symmetric about its angle bisector $\overrightarrow{AK}$.

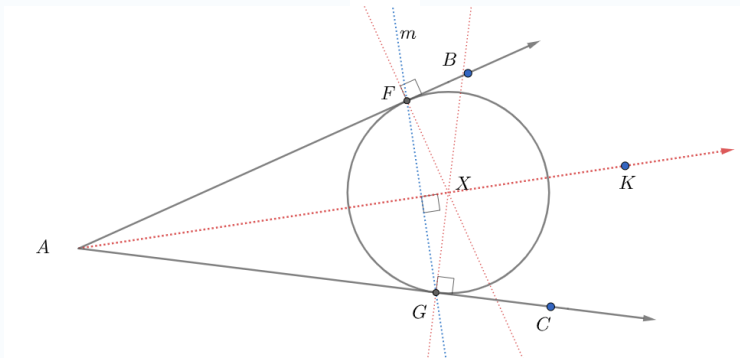


We now return to the main instruction sequence.

Student View: Slide 14

Let angle $\angle BAC$ be given with angle bisector $\overrightarrow{AK}$. Let Q be any point on $\overrightarrow{AK}$. Then Q is equidistant¹ from the sides of $\angle BAC$.

PROOF: Let m be any perpendicular through $\overrightarrow{AK}$ and say m intersects $\angle BAC$ at points F and G . Then $\overrightarrow{AK}$ is contained in the perpendicular bisector of $\overline{FG}$ by symmetry. Therefore, given any point on $\overrightarrow{AK}$, say X , $XF = XG$. Now take X suitably chosen on ray AK to establish the result.²

*Application**Angle Bisector Theorem*

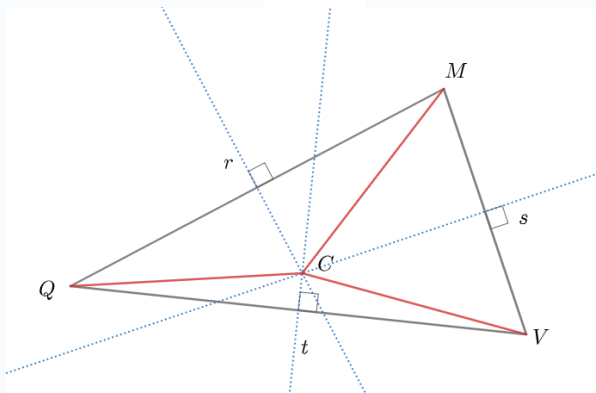
1. It must be emphasized that measurement of distance is along lines perpendicular to the sides of the angle. Any line perpendicular to the angle bisector is in fact "bisected" by it, but that is not the way distance is measured. Make sure to make this point clear.
2. If time permits, extend this discussion to account for why the angle bisectors of a triangle must be concurrent.

Student View: Slide 15

CHECK FOR UNDERSTANDING: How does the concept of symmetry help us understand WHY:

1. The perpendicular bisectors of the sides of a triangle are concurrent.
2. The angle bisectors of the angles of a triangle are concurrent.

EXPLANATION OF 1.



GIVEN $\triangle QMV$ with circumcenter C on lines r, s as shown.

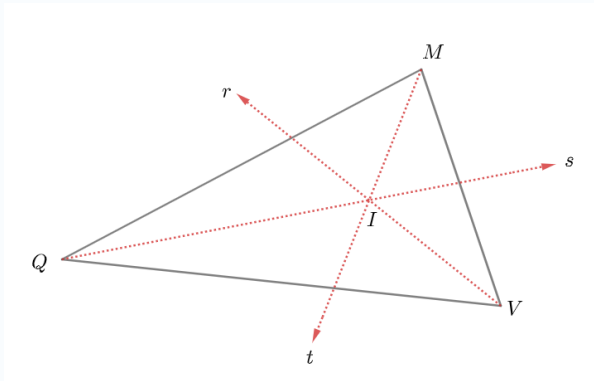
We need to explain WHY C also lies on t . Here is why. r is the perpendicular bisector of segment QM and C is on the intersection of r and s . Therefore $CQ = CM$. But C is also on s , the perpendicular bisector of segment MV , hence $CM = CV$ and it follows therefore that $CQ = CV$ since these are both equal to CM . Since $CQ = CV$, it follows that C must be equidistant from Q and V which means that C is on t , the perpendicular bisector of $\overline{QV}$. It follows that r, s, t must be concurrent at C .

Applications

Explaining Concurrencies

1. In this first discussion, note that the distances are computed from the circumcenter to the vertices of the triangle. Since the bisectors of the sides of the triangle are lines of symmetry, their incidence point forms three isosceles triangles with respect to the vertices of the ambient figure. This is the heart of the reasoning here.
2. In the second discussion the emphasis is on the distance between the alleged point of concurrence and the sides of the angles. Note that in both discussions, critical dependence is made on the concept of measure. In both of these discussions we are preparing students to be able to fashion arguments for themselves using reasoning skills based on previously established facts and assumed principles.

EXPLANATION OF 2.



GIVEN $\triangle QMV$ with incenter I on angle bisectors r, s as shown. We need to explain WHY I also lies on angle bisector t . Here is why. Since I lies on the angle bisector r ,

$$\text{dist}(I, \overline{MV}) = \text{dist}(I, \overline{QV}).$$

Now, since I also lies on the angle bisector s ,

$$\text{dist}(I, \overline{QM}) = \text{dist}(I, \overline{QV}).$$

Hence

$$\text{dist}(I, \overline{QM}) = \text{dist}(I, \overline{MV}).$$

And this means that I must lie on the angle bisector of $\angle QMV$ which of course is t .

Student View: Slide 16

We have done what we set out to do. 🧐

In this lesson we have covered a lot of material!

- We reviewed our basic constructions and learned more about constructing perpendicular and parallel lines.
- We introduced the formal term “postulate” that we will continue to use throughout this course.
- We discussed in detail the meaning of measurement.
- We introduced the concept of symmetry and its role in geometry.
- We concluded with clear explanations of why the concurrencies we saw in lessons 4 and 5 must be the case.

In our next lesson we will continue to apply these ideas and discuss further conclusions we can draw regarding angles, lines and structures associated with them.

Concluding Thoughts

iQ3R Mathematics — Geometry¹

Lesson 7 Module 1²

June 11, 2022

This 7th lesson presents a combination of materials from the New York Curriculum which are introduced in their lesson 6 and lesson 8. This is the first official departure from topic A and beginning of topic B materials. Students will review formerly learned geometry facts and practice citing the geometric justifications in anticipation of unknown angle proofs.

¹ Typically in the United States grade 9 students take either Algebra 1 or Geometry depending on the school and the performance of the student in 8th grade. Generally, Geometry is taken by students who are on a more accelerated track.

² Adapted for individualized instruction from the EngageNY curriculum for New York State public and private instructors.

Student View: Slide 1

Overview

IN THIS LESSON you will review some of the material you previously studied in earlier grades on angles. Specifically, the focus will be on solving for unknown angles — lines at a point and triangles. In each of the cases we will review how to find unknown angles.

1. When lines intersect at a point
2. When n angles lie on a line
3. When rays share a common vertex at a point
4. Angles in triangles and Exterior Angle Theorem
5. Applications

Overview

Lesson 7 Supported Standard

Common Core Focus Standards

There is only one supported standard throughout the topic B materials in this Module. That is:

G-CO.C.9

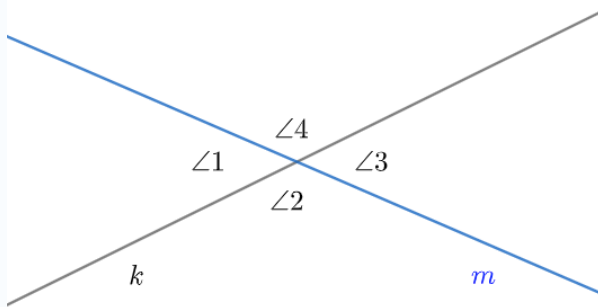
Prove theorems about lines and angles. Theorems include: vertical angles are congruent; when a transversal crosses parallel lines, alternate interior angles are congruent and corresponding angles are congruent; points on a perpendicular bisector of a line segment are exactly those equidistant from the segment's endpoints.

Student View: Slide 2

*Lines Intersecting at a Point**Review of Facts*

KNOWLEDGE POINTS: On two intersecting lines

Lines k and m intersect at point P (not shown). There are four angles with a common vertex at P . Angles 1 and 3 form a *vertical pair*, as do angles 2 and 4.

**FACTS:**

- $m\angle 1 = m\angle 3$
- $m\angle 2 = m\angle 4$

REASON: If two angles form a vertical pair, then they have equal measure.

ABBREVIATION: If $\angle A$ and $\angle B$ form a vertical pair, write “ A and B are vert. $\angle$ ’s.”

HOW TO STATE: (as a reason in a proof or exercise) If you need to state *why* $m\angle A$ and $m\angle B$ (vert. $\angle$ ’s) are equal, write “Vertical angles are equal in measure.”

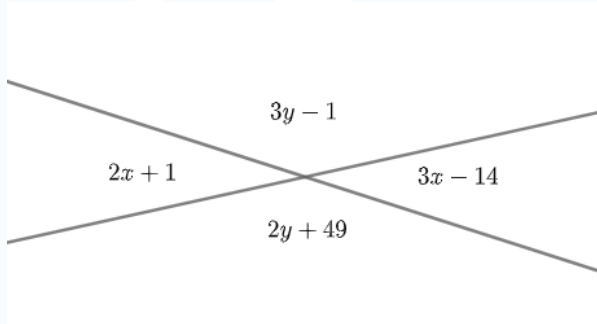
Instructor Notes

- We introduce the term “vertical pair” to help students understand that there is first a condition that must be met before they can simply say “vertical angles are equal in measure.” Some students think that just any angles having a common vertex are “vertical”.
- While there are only two vertical pairs, there are four “linear pairs” formed by two intersecting lines. They know about linear pairs so it is a good idea to ask them to find all the linear pairs here as well.
- They should be familiar with the abbreviations here, but make sure to review for completeness.
- For most of this course angle measure is always in degrees.

Student View: Slide 3

*Lines Intersecting at a Point**Practice*

In the diagram, both lines in the plane intersect. Each algebraic expression represents a degree measurement.[†]



1. Find the value of $x + y$. State your reasons.
2. What is the measure of either of the smaller angles?
3. What is the measure of either of the larger angles?
4. How could you have obtained the measure of a larger angle from the measure a smaller angle without finding y ?

Instructor Notes

[†] The alternative would be to write, for example, $(3y - 1)^\circ$, which we will do later. The author was a bit too hasty in putting the diagram together. Nevertheless, this is a common way of specifying the meaning of algebraic expressions in diagrams and should not be a source of much confusion. Simply, draw attention to the difference between, for example, the meaning of $3y - 1$ — representing a measurement of degrees, and the variable y — representing a generic real number.

1. $x + y = 65$. The larger angles are vert. $\angle$ s, as are the smaller. Therefore their measures are equal. We therefore have $2x + 1 = 3x - 14$. Solving, $x = 15$. Similarly, $3y - 1 = 2y + 49$ so that $y = 50$.
 2. Plug $x = 15$ into either x -expressions to get 31 for the smaller angles.
 3. Plug $y = 50$ into either y -expressions to get 149.
 4. A larger angle forms a linear pair with either of the smaller, so their angle sum is 180. Once we find that the smaller angle is 31, subtract from 180 to get the result.
- They have seen that linear pairs (angles on a line) are supplementary and the supplementary angles add to 180.
 - Since all measures of all angles in this course (for the most part) are in degrees, we will suppress always writing the degree symbol except when necessary to avoid confusion. Alert the student to this convention and note that they do not need to always supply a degree symbol either. (For this course)

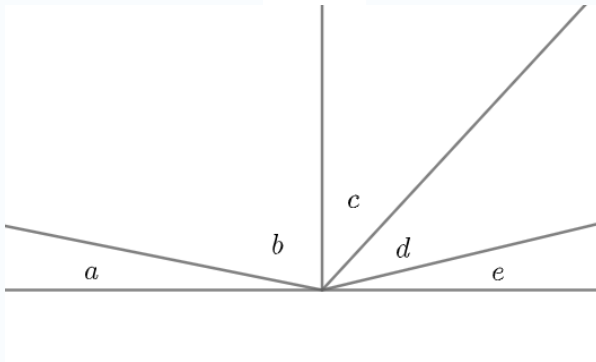
Student View: Slide 4

*Several angles on a line**Review of Facts*

KNOWLEDGE POINT: Consecutive adjacent[†] angles on a line
Suppose a sequence of n consecutive adjacent angles satisfies the two conditions.

- The interiors of the angles are all disjoint.
- The angle formed by the first $n - 1$ angles¹ and the last angle form a *linear pair*.

Then, the sum of all the angle measures is 180.



The figure above shows the case where $n = 5$ measurements (in degrees) indicated by the variables.

$$a + b + c + d + e = 180$$

ABBREVIATION: $\angle$ s on a line

HOW TO SAY²: Consecutive adjacent angles on a line sum to 180.

Instructor Notes

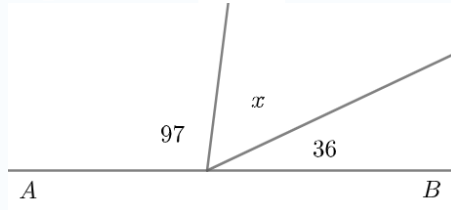
[†] This concept was defined earlier make sure to clarify that the student understands this.

1. Well, this is not technically correct. Since an angle is a union of rays, we should say “the angle formed by the union of the first $n - 1$ angles.” But we will go with the abuse of language to avoid being even more cumbersome.
2. We are deliberately omitting all the extra verbiage of the last section. We hope the point is understood.

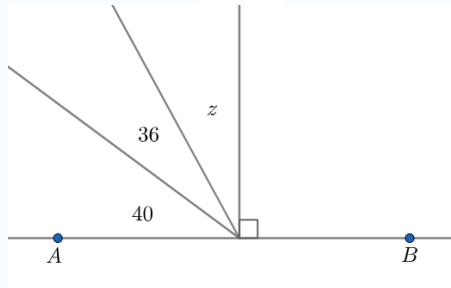
Student View: Slide 5

*Several angles on a line**Practice*[†]

EXERCISE 1. Find the value of x . All angles are on line AB .
State your reasons.



EXERCISE 2. Find the value of z . All angles are on line AB .
State your reasons.

*Instructor Notes*

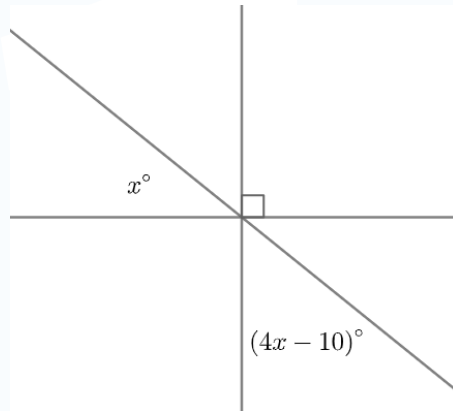
[†] They are also familiar with the “angle addition” postulate, as well as fact linear pairs are supplementary. We use these facts in this practice implicitly.

1. $x = 47$. Consecutive adjacent angles on a line sum to 180.
2. $z = 14$. Consecutive adjacent angles on a line sum to 180.

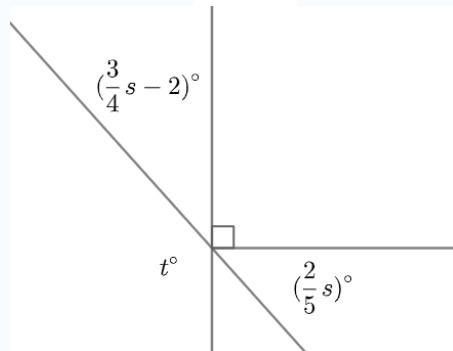
Student View: Slide 6

*Several angles on a line**Practice*

EXERCISE 3. Find the value of x . State your reasons.



EXERCISE 4. Find the value of s and t . State your reasons.

*Instructor Notes*

⇒ Note here the use of the degree symbol $^\circ$. We are using that here to differentiate the algebraic expression that results in a degree measurement. So here the degree symbol serves to differentiate between the values of x which are real numbers, and the degree measures yielded by functions of x .

1. $x = 20$. Vertical angles are equal in measure. Consecutive adjacent angles on a line sum to 180.
2. $s = 80$ and $t = 122$. Consecutive adjacent angles on a line sum to 180.

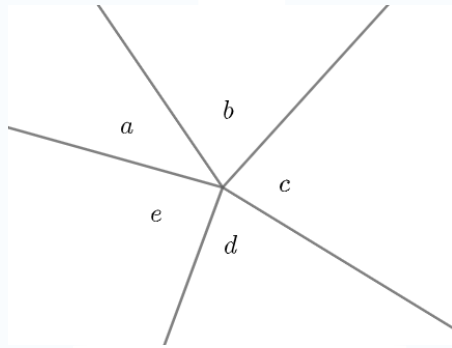
Student View: Slide 7

*Angles at a point**Review of facts*

KNOWLEDGE POINT: Suppose three or more rays all have the same vertex and their interiors do not overlap. Then sum of the measure of all angles formed is 360.

ABBREVIATION: $\angle$ s at a point

HOW TO SAY: Angles at a point sum to 360.



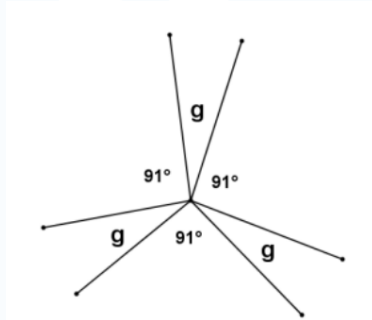
In the figure there are five measurements* indicated.

$$a + b + c + d + e = 360.$$

Instructor Notes

* We again abuse notation slightly. Sometimes when we mean angle a we will write simply a in the drawing. For this reason we said that “measurements” are represented here. This sort of context dependent abuse of notation is common in textbooks and students should get used to using “context” clues to determine how to interpret the labels. A similar problem happens in algebra when we have to make the distinction between $f(x)$ — the value of the function f at x , and the function f . Nevertheless we will try to be as consistent as we can.

Student View: Slide 8

*Angles at a point**Practice*

EXERCISE 1: In the figure above, g represents the *number of degrees* in each of the three congruent angles. Find the value of g . State your reasons.

Instructor Notes

SOLUTION:

Angles at a point sum to 360.

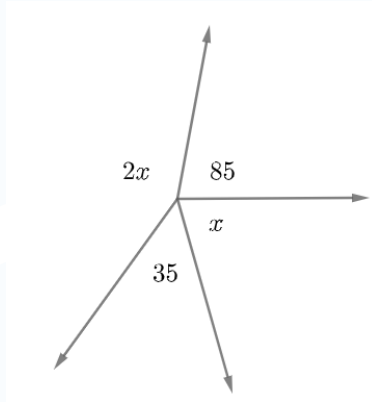
$$g = 29.$$

Did you notice what happened? The style of the figures changed. The figure you are seeing in this slide is a screenshot from the New York material. I am experimenting to see how much less time it takes if we use this method. Up until this point, I have been either designing my own figures in Geogebra or LaTeX and changing some of the features while still retaining the main idea. (For the most part)

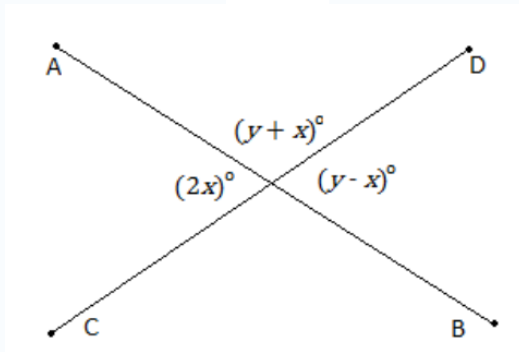
OBSERVATIONS:

- 1° This might violate copyright laws if the exact figures and problems are included in published work, so eventually similar problems will have to be crafted and coded.
- 2° There are many conceptual issues with the actual New York material as it actually stands. This figure is a great example!
- 3° Note that in their convention g is a *label* for an angle. In fact when the present the solution they even write $m\angle g = 29^\circ$. There is a problem here. Which angle g ? Those three angles are congruent but they are not *the same* angle.
- 4° In my convention, we would say that g is the angle measure in degrees and leave off the little $^\circ$ symbol on the known angles.
- 5° As we move forward we need to take note of these details.
- 6° The time we save using screenshots might be offset by the time we have to spend modifying the problem to use the screenshot! 🖨️🔍

Student View: Slide 9

*Angles at a point**Practice*

EXERCISE 2: In the figure above, all rays have a common vertex. Each expression represents *how many*[†] degrees are in each angle measurement. Find the value of x . State all of the reasons you use.



EXERCISE 3: The figure above shows intersecting segments $\overline{AB}$ and $\overline{CD}$. The angle measures^{††} are given here in terms of algebraic expressions. Find x and y . State the reasons for your answers.

Instructor Notes

† Again we are using this convention that we will use the language of the problem to avoid confusion. Be sure to point this out. So x is a number of degrees as is 35.

†† The difference here is this: say E is the intersection point. Then we would properly write $m\angle AED = (y + x)^\circ$. But if we hold to convention that angle measure is ALWAYS in degrees, unless otherwise stated, we could just as well write $m\angle AED = y + x$. This is why the degree symbol is a bit redundant.

SOLUTIONS:

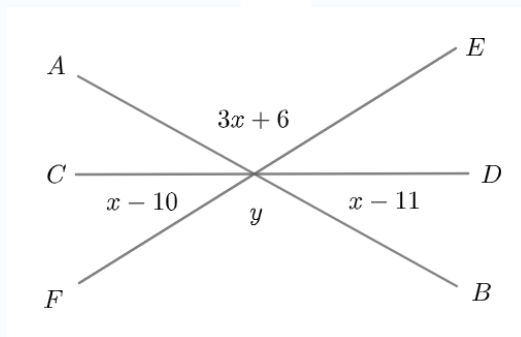
EXERCISE 2: $x = 80$. Angles at a point sum to 360.

EXERCISE 3: $x = 30$ and $y = 90$. Vertical angles are equal in measure. Angles at a point sum to 360.

Student View: Slide 10

*Angles at a point**Practice*

EXERCISE 4: In the figure below, segments $\overline{AB}$, $\overline{CD}$, and $\overline{FE}$ are all concurrent[†]. Each of the algebraic expressions represents how many degrees are in each angle measurement. Find the values of x and y and state all the reasons you to find the answer.

*Instructor Notes*

† This is a great chance to review this topic from the last couple lessons. Three or more lines (or segments) are concurrent when they share the same point of intersection.

Why is do we need labeled segments?

SOLUTION: $x = 39$ and $y = 123$. Vertical angles are equal in measure. Angles at a point sum to 360.

Do we need to show the algebra?

Student View: Slide 11*Angles in a triangles**Review of facts*

KNOWLEDGE POINT: The sum of the measures of the angles in any triangle is always 180.

EXPLANATION: We will chart out a series of steps you can go through on your own to verify this.

Instructor Notes

This is an extremely widely used result. It is important that students have some inclination of how it is related to work we have done so far. The “explanation” we provide here is not a rigorous proof, but merely a way of connecting this familiar idea to the most recent review exercises.

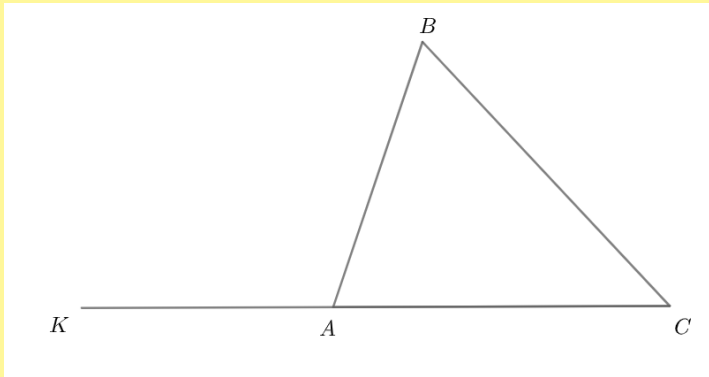
An animation would be helpful.

Lesson Design Specifications (Please Read)

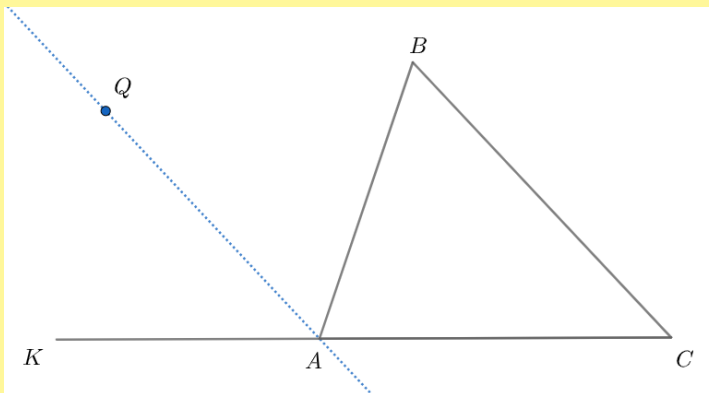
Guided Practice:

At this point student and teacher will do one of the following:

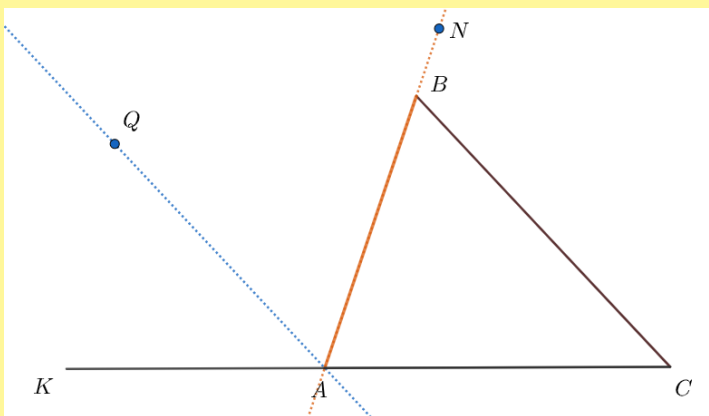
- Navigate to Geogebra on a shared screen **OR**
- Launch an app that opens up a shared window for interactive geometry **OR**
- Share a whiteboard with appropriate drawing tools.



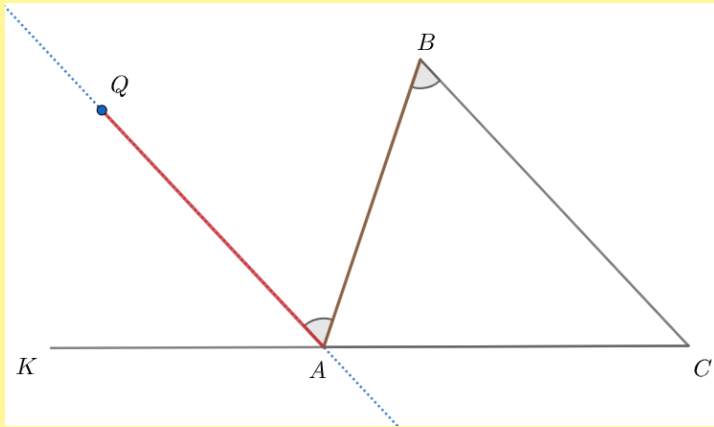
STEP1: Start with $\triangle ABC$ and extend segment AC as shown.



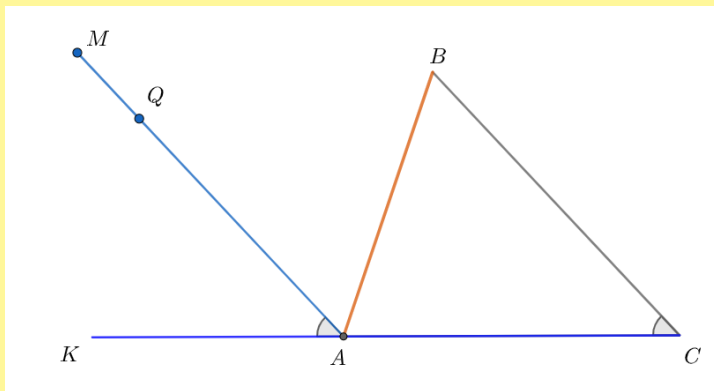
STEP 2: Draw line AQ parallel to $\overline{BC}$ as shown.



STEP 3: Locate N on the same side of line AQ as B so that $\angle QAN \cong \angle CBA$. Observe: B lies on line AN . (We can do this for example by verifying that $AN = AB + BN$.)



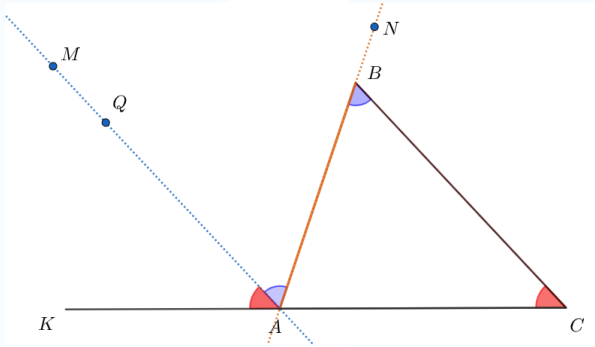
STEP 4: It follows that $\angle KAQ$, $\angle QAB$ and $\angle BAC$ are all angles on a line. (We needed to establish that B was in fact on line AN in order to make this statement.) Therefore the sum of their measures is 180. We will now show that $\angle KAQ \cong \angle ACB$.



STEP 5: To verify that $\angle KAQ \cong \angle ACB$ we follow a similar procedure as in **STEP 3**. Locate M on the same side of $\overline{CK}$ as B so that $\angle MAK \cong \angle BCA$. Observe that Q lies on segment AM .

We now return to the main instructional sequence.

Student View: Slide 12

*Angles in a triangle**Review of facts*

SUMMARY: We have shown that $\angle MAK$, $\angle QAB$ and $\angle BAC$ are all[†] angles on a line and therefore the sum of their measures is 180. Since $\angle MAK \cong \angle BCA$, and $\angle QAB \cong \angle CBA$ we can conclude that the sum of the measures of angles $\angle BCA$, $\angle CBA$, and $\angle BAC$ is also 180! These are the angles in the triangle.

In our next lesson, we will show how to give a much simpler explanation without copying angles.

Instructor Notes

[†] Actually not quite. We showed $\angle QAK$, $\angle QAB$ and $\angle BAC$ were angles on a line.

Since Q is on $\overleftrightarrow{AM}$ the statement we use here is valid. Some clever students might comment on this.

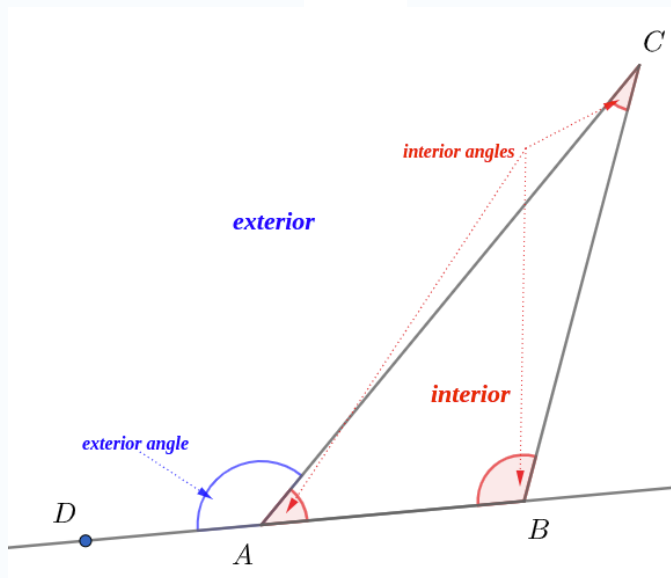
Student View: Slide 13

*Angles in a triangle**Review of facts*

VOCABULARY: Let $\mathcal{T}$ be a given triangle in the plane.

- 1° If at least two sides of $\mathcal{T}$ are congruent, $\mathcal{T}$ is said to be *isosceles*.
- 2° If all three sides of $\mathcal{T}$ are congruent, then $\mathcal{T}$ is said to be *equilateral*.
- 3° The *interior* of a triangle is the intersection of the interiors of each of its angles which are called *interior angles*.
- 4° A point is on the *exterior*[†] of a triangle if it is any point not in the interior or on its sides.
- 5° $\angle DAC$ is called an *exterior angle*^{††} if D is on line AB and $\angle BAC$ and $\angle DAC$ form a linear pair.

The figure below should help visualize 3°, 4° and 5°.



In the figure, $\angle B$ and $\angle C$ are said to be *remote interior angles* relative to the exterior angle $\angle DAC$. *

Instructor Notes

These vocabulary terms should be familiar. We just review them here for completeness.

† A triangle divides the plane into two disjoint sets: one bounded, the other unbounded. The bounded region is convex, while the unbounded region is not. An easy, but useful observation.

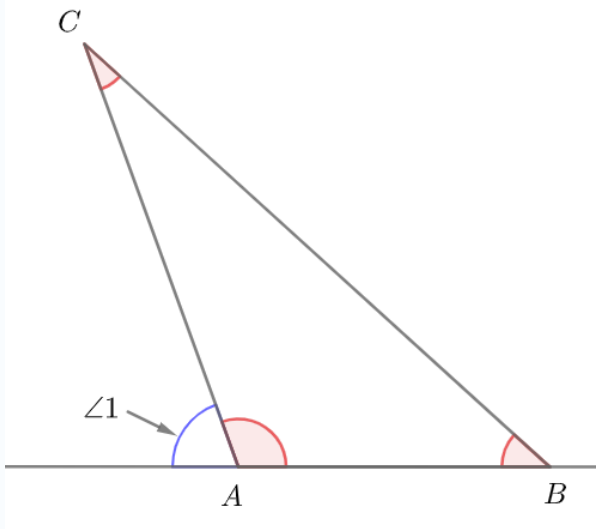
†† This forces the condition that A must be between D and B . A good observation to point out. What goes wrong if that condition is violated?

* The exterior angle $\angle KAB$ formed along side AC also shares $\angle B$ and $\angle C$ as remote interior angles, so it is important to specify which exterior angle we are speaking of. Their measures however are the same.

Student View: Slide 14

*Angles in triangles**Review of facts***KNOWLEDGE POINT:** *Exterior Angle Theorem*

In any triangle the measure of an exterior angle is always the sum of its two remote interior angles.



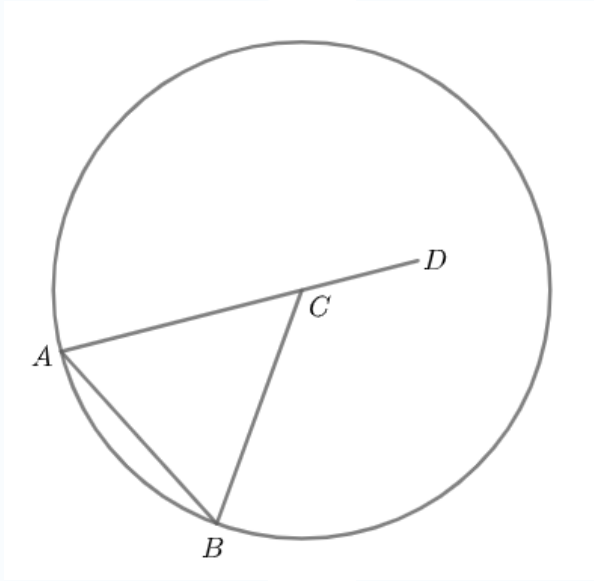
Here, according to the figure, by the Exterior Angle Theorem we can conclude that

$$m\angle 1 = m\angle B + m\angle C.$$

Instructor Notes

- It is fine to label angles like this when there is no ambiguity.
- $\angle 1$ and $\angle A$ form a linear pair, so there should be no confusion here.
- Later in the course, we will give a proof of this fact. We just accept it as a given at this point to work with it.

Student View: Slide 15

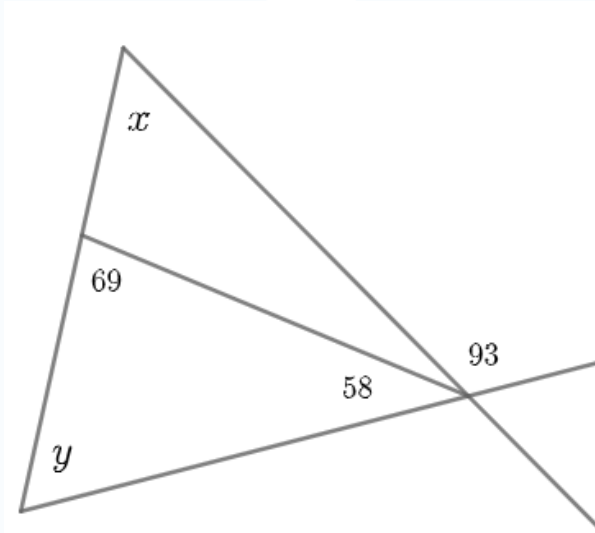
*Applications**Relationship with circles*

In the figure, circle C has center at C . A and B lie on circle C . $\overline{AD}$ is part of the diameter and AD is larger than the radius of circle C . If $m\angle DCA = 122$ find the measures of all the interior angles of $\triangle ABC$. State the reasons for each of your calculations. (Hint: the base angles of an isosceles triangle are congruent.)

Instructor Notes

- 1° $\angle DCB$ is exterior to $\triangle ABC$ (Definition)
- 2° $m\angle DCB = m\angle A + m\angle B$. (Exterior angle theorem.)
- 3° $m\angle A = m\angle B$. (Hint, and cong. $\angle$ s have the same measure)
- 4° $2m\angle A = m\angle DCB = 122$. (Substitution and given)
- 5° $m\angle A = 61 = m\angle B$ (Division)
- 6° $m\angle BCA = 58$ (Either sum of angles in a triangle or supplementary angles.)

Student View: Slide 16

*Applications**Relationship with internal triangles*

In the figure above, all segments which appear to intersect do intersect. All algebraic and numeric quantities represent how many degrees are in the angle measurement. Find x and y . State your reasons for your calculations.

Instructor Notes

$$1^\circ \quad 93 = x + y \text{ (Exterior angle theorem)}$$

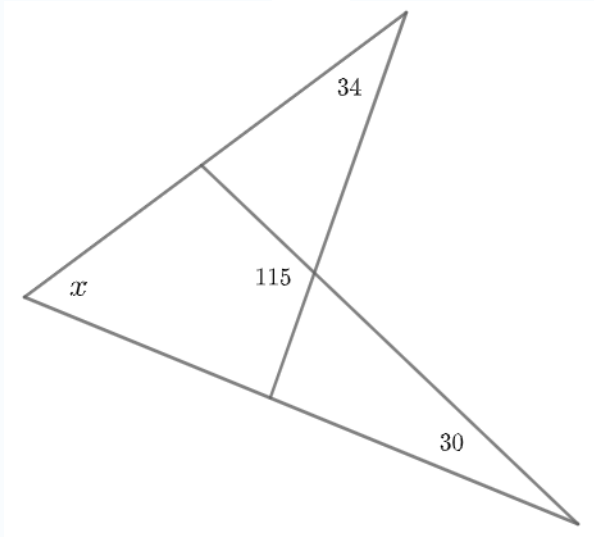
$$2^\circ \quad y + 69 + 58 = 180 \text{ (Sum of angles in a triangle)}$$

$$3^\circ \quad y = 53, x = 40 \text{ (Arithmetic and } 1^\circ)$$

Student View: Slide 17

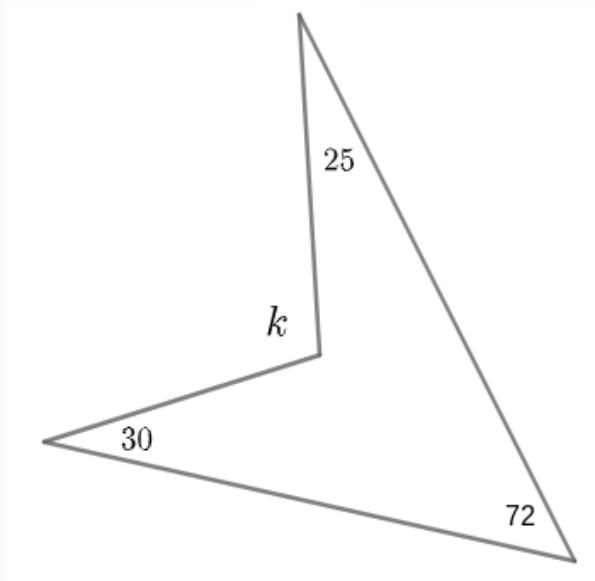
*Applications**Overlapping triangles*

In the figure both large triangles “overlap”. Segments which appear to intersect really do. The algebraic and numeric quantities represent the number of degrees in the angle measured. Find x . State your reasons.

*Instructor Notes*

1. $x = 51$.
2. The measures of supplementary angles add to 180.
3. Sum of the measures of the angles in a triangle.

Student View: Slide 18

*Applications**Angle chasing and “helping” lines*

In the figure both large triangles “overlap”. Segments which appear to intersect really do. The algebraic and numeric quantities represent the number of degrees in the angle measured. Find k . State your reasons.

Instructor Notes

1. $k = 127$.
2. Draw helping lines to make the figure look like the one in Slide 17.
3. Use the same reasoning in reverse to find k .